



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0370



Agricultural Machinery – Biomass Shredder WEIFANG 4" INLET

Test Report Number	2018 – 0370
Valid Until	September 11, 2018
Agricultural Machinery	Biomass Shredder
Type	Cart type, Closed cylinder, Horizontal blade,
Brand	WEIFANG
Model	4" INLET
Test Requested by	SOUTHLAKE CONSTRUCTION AND DEVELOPMENT CORPORATION Unit 33/6 Francisco Canicosa Ave., Sta. Rosa Commercial Complex, Brgy. Balibago, Sta. Rosa City, Laguna

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 244:2010	AMTEC Test
Input capacity, kg/h	400 ^a	334.6
Noise level, dB(A), maximum	96*	100.8
Shredding efficiency, %, minimum	90	100

Note: *Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines. 1983.

^aManufacturer's specifications

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous valid test reports of similar brand and model	2018 – 0366
3. Prime mover used	5.7 kW CHINA BRAND Diesel Engine
4. Manufacturer	SHANDONG WEIFANG LUZHONG TRACTOR CO., LTD. Caihong Road, Weifang City, Shandong Province, China
5. Loading hopper	Accessible to the operator
6. Operation	Recommended shredding chamber shaft speed should be attained before feeding. Proper size of input materials should be fed.
7. Adjusting parts	None
8. Repairing of parts	Not observed
9. Collecting output	Output was collected after operation.
10. Cleaning parts	Not observed
11. Safety	Provided with warning stickers. Transmission system has safety cover.
12. Availability of the switches needed	Available
13. Transporting the machine	With cart wheels and transport handle
14. Number of operator/s	2 (Feeding operators)
15. Other remarks	No breakdown occurred during operation.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	June 28, 2018
Location of Test	AMTEC Building, Pili Drive, University of the Philippines Los Baños
Items Tested	Input capacity, Shredding efficiency, Noise level, Speed of rotating components, Fuel consumption, and Number of operators

METHODS OF TEST

The biomass shredder was tested based on the PNS/PAES 245:2010 Agricultural Machinery – Biomass Shredder – Methods of Test.

DESCRIPTION OF THE MACHINE

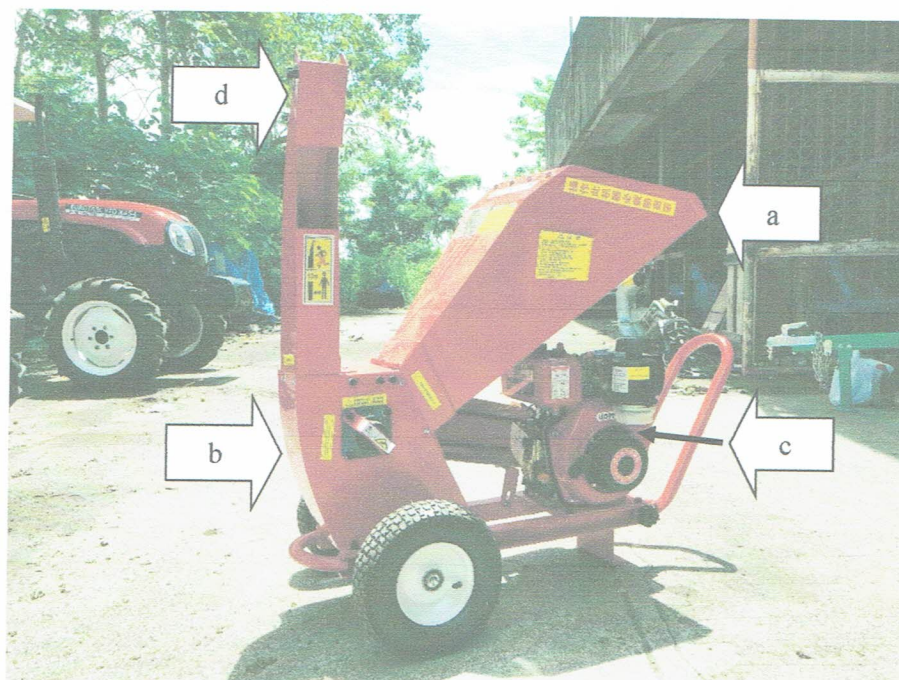


Figure 1. Parts of the WEIFANG 4" INLET Biomass Shredder powered by 5.7 kW Diesel Engine
(a) Hopper, (b) Shredding Chamber, (c) Prime mover, and (d) Output chute

SPECIFICATIONS

Item		Manufacturer's Specifications ^a	AMTEC Verification ^b
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	ND	1230
1.1.2	Width	ND	600
1.1.3	Height	ND	1085
1.2	Weight without primemover, kg	ND	NM
2	Loading hopper		
2.1	Material	ND	2.0 mm G.I. sheet
2.2	Bottom opening, mm		
2.2.1	Length	ND	170
2.2.2	Width	ND	150
2.3	Height from the ground, mm	ND	905
2.4	Other features	ND	Has rubber guard at the inlet
3	Shredder assembly		
3.1	Blade cover		
3.1.1	Type	ND	Semi-circle
3.1.2	Materials	ND	5.95 mm G.I. plate
3.2	Blade		
3.2.1	Material	ND	G.I. plate
3.2.2	Type	ND	Drum
3.2.3	Dimensions, mm		
3.2.3.1	Length	ND	150
3.2.3.2	Width	ND	60
3.2.3.3	Thickness	ND	9.32
3.2.4	Shape	ND	Rectangular
3.2.5	Number of blades	ND	2
3.2.6	Clearance to the shredding cylinder, mm	ND	NM

Note: ^aManufacturer did not submit the complete machine's specification.

^bBased on actual performance results, on-site measurements

ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specifications ^a	AMTEC Verification ^b
4	Output chute		
4.1	Material	ND	2 mm GI sheet
4.2	Dimensions, mm		
4.2.1	Length	ND	300
4.2.2	Width	ND	83
4.3	Height from the ground	ND	1055
4.2	Other features	ND	A deflector is installed at the end of the outlet.
5	Safety devices	ND	Has transmission guard
6	Special features	ND	None
7	Primemover		
7.1	Internal Combustion Engine		
7.1.1	Brand	ND	ND*
7.1.2	Model	ND	ND*
7.1.3	Make or manufacturer	ND	China ^c
7.1.4	Serial number	ND	ND
7.1.5	Type	ND	Four-stroke, single vertical cylinder, diesel engine
7.1.6	Rated power, kW	ND	5.70 ^c
7.1.7	Rated speed, rpm	ND	2600 ^c
7.1.8	Cooling system	ND	Air-cooled
7.1.9	Starting system	ND	Rope recoil
7.1.10	Weight, kg	ND	70 ^c

Note: ^aManufacturer did not submit the complete machine's specification.

^bBased on actual performance results, on-site measurements

^cBased on engine's nameplate

*Written in Chinese characters

ND – No Data

SPECIFICATIONS (Cont’n)

Item		Manufacturer's Specifications ^a	AMTEC Verification ^b
8	Body		
8.1	Material	ND	5.95 mm G.I. plate
9	Chassis		
9.1	Material	ND	C-bar
9.2	Size, (L x W x t), mm	ND	50 x 40 x 6.29
10	Transmission system		
10.1	Type	ND	Belt and pulley (A-1499)
10.2	Engine to shredder shaft	ND	76 mm x 2A
10.3	Speed reduction ratio	ND	1:1.5

Note: ^aManufacturer did not submit the complete machine's specification.

^bBased on actual performance results, on-site measurements

ND – No Data

RESULTS

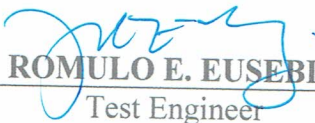
Item		AMTEC Test	
1	Conditions of test materials		
1.1	Type	Assorted wood branches	
1.2	Variety	Unknown	
1.3	Source	Laguna	
1.4	Weight, kg	22.75	
1.5	Moisture content, % _{wb}	25.1	
1.6	Dimensions, mm		
1.6.1	Length	1412	
1.6.2	Width	20	
2	Weight of input, kg	22.75	
3	Weight of output, kg	NM	
4	Operating time, h	0.068	
5	Input capacity, kg/h	334.6	
6	Shredding efficiency, %	100	
7	Speed of components, rpm	Without load	With load
7.1	Prime mover	3426	3312
7.2	Shredding shaft	2479	2392
8	Noise Level, dB(A)		
8.1	Feeding Operator	100.8	97.6
9	Fuel consumption		
9.1	Fuel consumed, mL	330	
9.2	Fuel consumption rate, L/h	0.93	
10	Number of operators	2 (Feeding operator)	

Note: NM – Not Measured

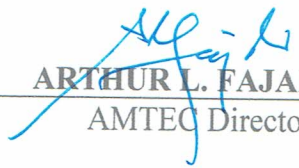
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JUN 28 2019
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